

7 Lab 6: MLTP

7.1 Introduction to the MLTP Simulation Framework

The Minimum Lap Time Problem (MLTP) is formulated as an Optimal Control Problem (OCP) to simultaneously find the optimal racing line and the corresponding control inputs (throttle, braking, and steering). The continuous-time problem is transcribed into a discrete Non-Linear Programming (NLP) problem using direct orthogonal collocation. The software framework CasADi is employed for algorithmic differentiation, while the IPOPT solver, utilizing an interior-point method, is used to find the optimal solution.

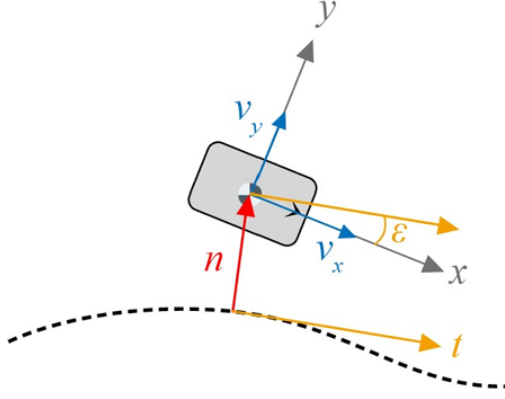


Figure 2: Motion of the vehicle along the track

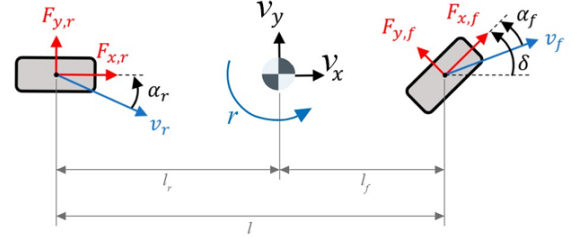


Figure 3: Single track model

The vehicle is modeled using a single-track (bicycle) approximation that includes non-linear tire forces computed via Pacejka's Magic Formula, longitudinal load transfer, and aerodynamic drag and downforce. To simplify the OCP discretisation, the independent variable is changed from time to traveled distance (s), allowing the track curvature and limits to be evaluated explicitly at each spatial node. The optimization is subject to several path constraints, including the friction ellipse, maximum motor power, and the prohibition of simultaneous throttle and brake application.

7.2 Objective Function and Regularization Penalties

To fully understand the numerical behavior of the optimizer, it is crucial to analyze the mathematical formulation of the Optimal Control Problem. The NLP solver (IPOPT) does not strictly minimize the lap time alone; rather, it minimizes a composite Objective Function (or Cost Function, J).

According to the simulation framework's architecture, the objective function includes the integration of the lap time alongside several regularization terms designed to penalize non-smooth control inputs:

$$J = \int_0^{s_f} S_f(w) ds + \sum (r_u U_k^2 + r_{du} \Delta U_k^2 + r_{du2} \Delta^2 U_k^2 + \dots) \quad (48)$$

Where U_k represents the control inputs (throttle, braking, steering) and ΔU_k represents their rate of change.

The regularization weights (r_{du} , r_{du2}) intentionally penalize aggressive, "bang-bang" control strategies to ensure realistic actuator behavior and numerical stability. However, this introduces a mathematical trade-off: a highly aggressive lap yields the minimum time but incurs a massive regularization penalty.

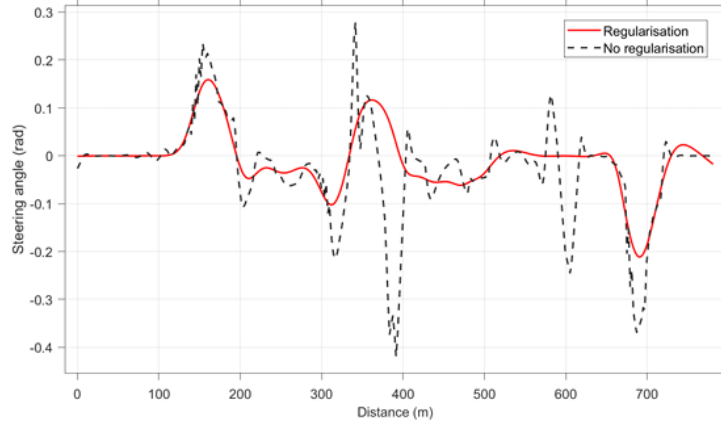


Figure 121 Example taken from the manual

7.3 Energy vs. Lap Time: The Pareto Front and Solver Anomalies

A sensitivity analysis is conducted by varying the maximum allowed energy consumption per lap. Simulations are run with energy limits ranging from 8 MJ to 20 MJ on the Barcelona circuit to evaluate the trade-off between lap time and energy deployment.

As the energy limit increases, the lap time decreases asymptotically until the vehicle reaches its physical limits (tire grip, aerodynamic drag, and maximum powertrain output).

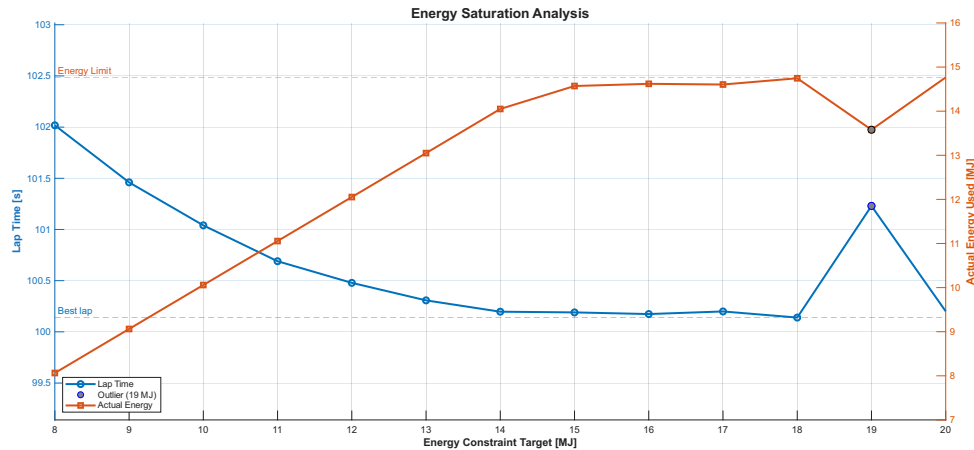


Figure 122 Lap time vs. Energy consumption limit.

Moving forward, the 15 MJ configuration will be adopted as the baseline. This value is chosen because it closely approaches the absolute performance limit while ensuring that the energy constraint does not become excessively loose. An anomalous result is observed for the 19 MJ configuration, whose lap time is approximately 1 s higher than the 15 MJ

baseline despite the energy constraint being inactive. One possible explanation is that the solver converged to a sub-optimal local solution influenced by the regularization terms included in the objective function:

$$J = \int_0^{s_f} S_f(w) ds + \sum (r_u U_k^2 + r_{du} \Delta U_k^2 + r_{du2} \Delta^2 U_k^2 + \dots) \quad (49)$$

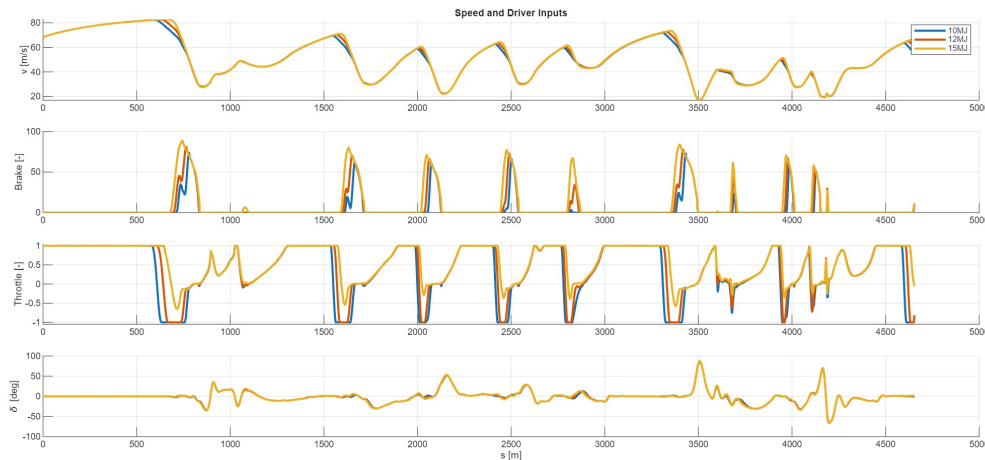
As shown in Figure ??, the 19 MJ solution presents smoother and more extended braking inputs, with regenerative braking reaching its saturation limit. This behaviour may reduce the quadratic penalties associated with rapid control variations, although at the cost of a longer deceleration phase and a higher lap time.

Since the energy constraint is inactive, the broader feasible region may have allowed IPOPT to converge towards this smoother solution. The resulting low energy consumption of 13.57 MJ could therefore be a secondary consequence of the reduced aerodynamic drag and limited use of mechanical braking. However, additional solver diagnostics and different initial guesses would be required to confirm this interpretation.

7.4 Telemetry Analysis and Lift & Coast Strategy

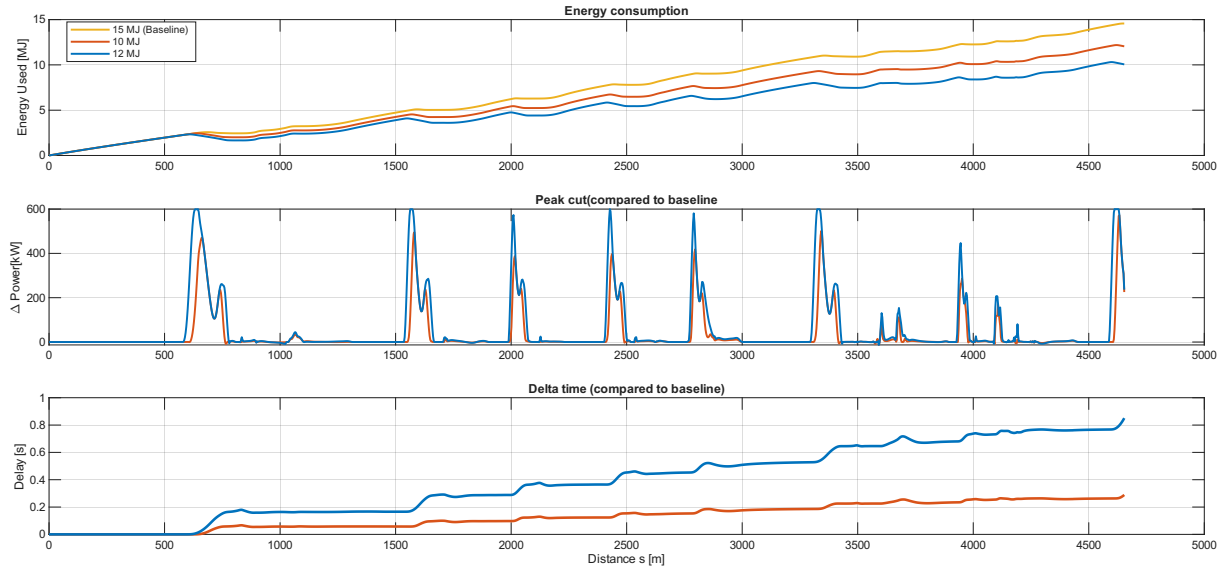
To understand the impact of energy constraints on driving behavior, the 10 MJ, 12 MJ, and 15 MJ (unconstrained) simulations are compared. The telemetry highlights the implementation of a "Lift and Coast" strategy by the optimal controller to satisfy the stringent energy limits.

Instead of applying conventional mechanical brakes, which dissipate kinetic energy as heat, the optimizer strategically cuts the throttle well before the braking zones of mid-to-low speed corners. This allows the vehicle to coast, maintaining momentum while saving significant amounts of energy. Furthermore, the telemetry reveals extensive use of regenerative braking. The negative torque capability of the electric motors is fully exploited during deceleration phases to recover energy, reducing the reliance on the mechanical braking system.



The severity of the energy constraint directly influences the activation of these strategies.

At 12 MJ, lift and coast is selectively applied only before the heaviest braking zones, where the $\Delta\text{Time}/\Delta\text{Energy}$ trade-off is most favorable. At 10 MJ, the budget is so restricted that coasting extends to high-speed sections and corner exits.



7.5 Spatial Distribution of Dominant Physical Limits

To verify that the optimizer correctly exploits the vehicle's performance envelope, the 15 MJ unconstrained lap is segmented based on the dominant physical limit at each spatial node.

The resulting track map demonstrates a precise transition between different dynamic states. In heavy braking zones and corner entries, the vehicle is strictly limited by front axle grip. During corner exits, as longitudinal acceleration combines with lateral forces, the rear axle grip dictates the maximum performance. On straight segments, the limitation shifts entirely to the powertrain: initially constrained by the maximum torque output of the electric motors, and subsequently by the constant power limit as speed increases.

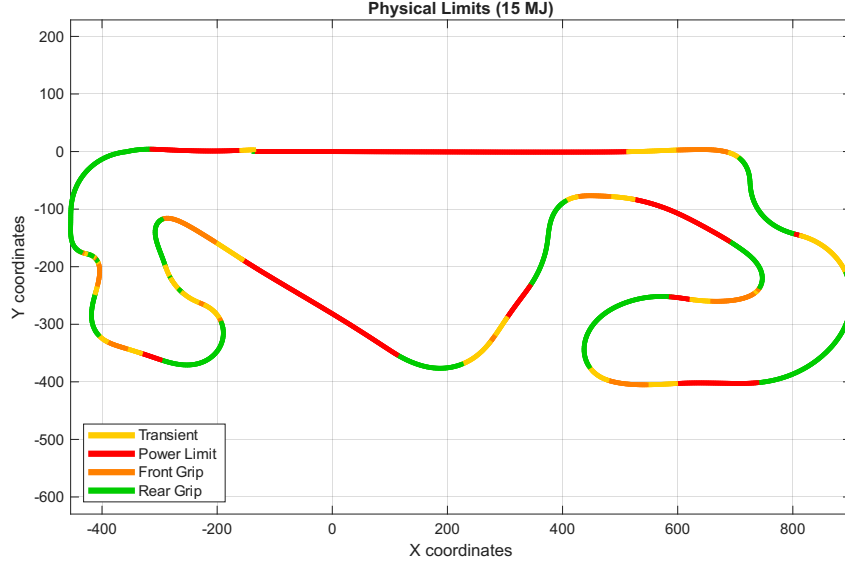


Figure 123

7.6 Energy Deployment and Savings Mapping

A final spatial analysis is performed to isolate the exact track sectors where energy conservation occurs. By calculating the instantaneous power difference (ΔP) between the 15 MJ and the constrained laps, a heat map of the circuit is generated.

The spatial distribution confirms that energy savings are not applied uniformly. The optimizer targets specific deceleration events, creating intense peaks of power reduction immediately preceding the geometric apexes. The comparison between the 12 MJ and 10 MJ heat maps visually demonstrates the non-linear expansion of the lift and coast zones required to meet increasingly severe energy targets.

To rigorously quantify the required driving style adaptations, the energy-saving strategy is evaluated through two distinct metrics: the spatial extension of the strategy deviation (meters) and its average intensity (ΔP in kW).

It is crucial to note that the 15 MJ baseline naturally includes regeneration phases during optimal braking. Therefore, the constrained configurations save energy not merely by extending coasting distances, but primarily by deepening the power cut within the existing deceleration zones.

The spatial analysis reveals a two-step optimization behavior. To achieve the 12 MJ target, the solver predominantly increases the intensity of the power cut in the same braking zones utilized by the baseline, yielding an average power reduction of **175.3 kW** over a modified track distance of **864.1 meters**. However, to meet the severe 10 MJ target, the optimizer exhausts the capacity to deepen the cut (saturating the maximum regeneration limits) **213.8 kW** modified strategy along **1230.2 meters**, actively encroaching onto the high-speed straights where the $\Delta \text{Time}/\Delta \text{Energy}$ trade-off is significantly more penalizing.

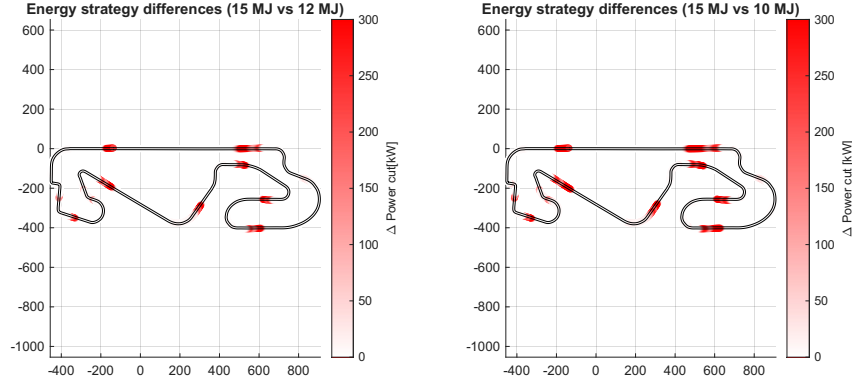


Figure 124

The G-G diagram illustrates the vehicle's operational envelopes across three different energy budgets (15 MJ, 12 MJ, and 10 MJ). As expected, the most significant deviations occur along the longitudinal axis (a_x). Under stricter energy constraints (e.g., 10 MJ), the solver drastically curtails maximum positive acceleration (traction) to respect the energy limit. Consequently, the vehicle achieves lower top speeds at the end of the straights, which in turn reduces the peak deceleration (braking) required, shrinking the lower boundary of the envelope. (Note: Raw acceleration output data is divided by 9.81 to properly represent 'g' units). Conversely, the lateral acceleration limits (a_y) remain virtually identical across all scenarios. This demonstrates that mid-corner performance is predominantly grip-limited rather than energy-limited; the solver consistently maximizes lateral limits to carry as much speed through the corners as the tires will allow.

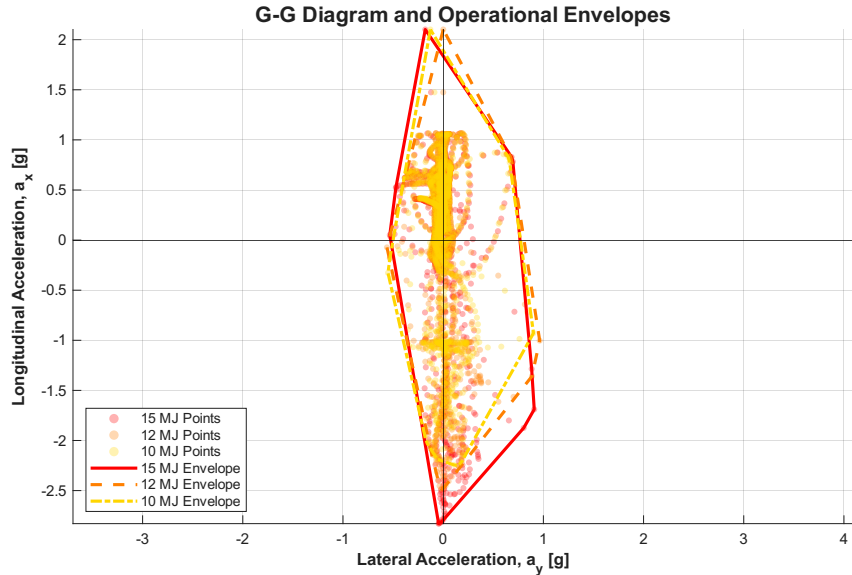


Figure 125

Interestingly, plotting the actual spatial trajectories revealed that the optimal racing line remains unchanged across the different energy modes. To meet the strict energy constraints, the optimal control strategy exclusively alters the longitudinal speed profile (e.g., modulating throttle, lowering top speeds, or coasting) rather than forcing the vehicle to take a different geometric path on the track.

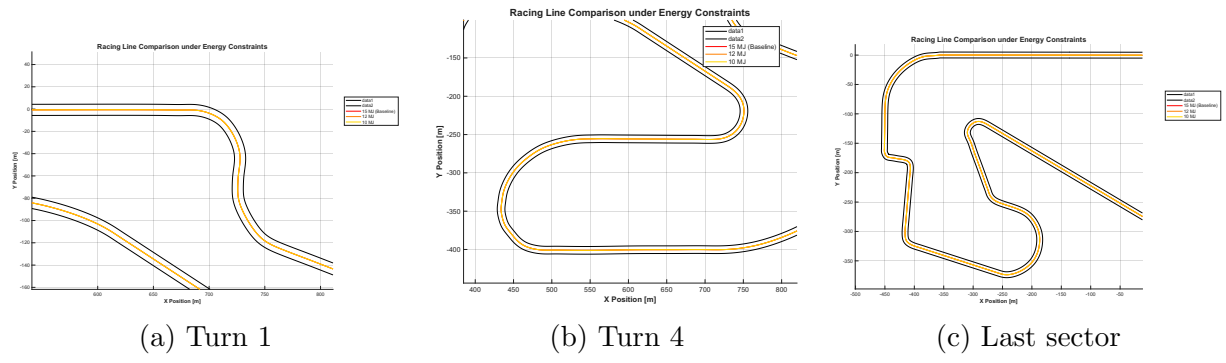


Figure 126Trajectory comparison .